LAB EXPERIMENT RECORD

Set 23: Employee Performance Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursaday

AIM

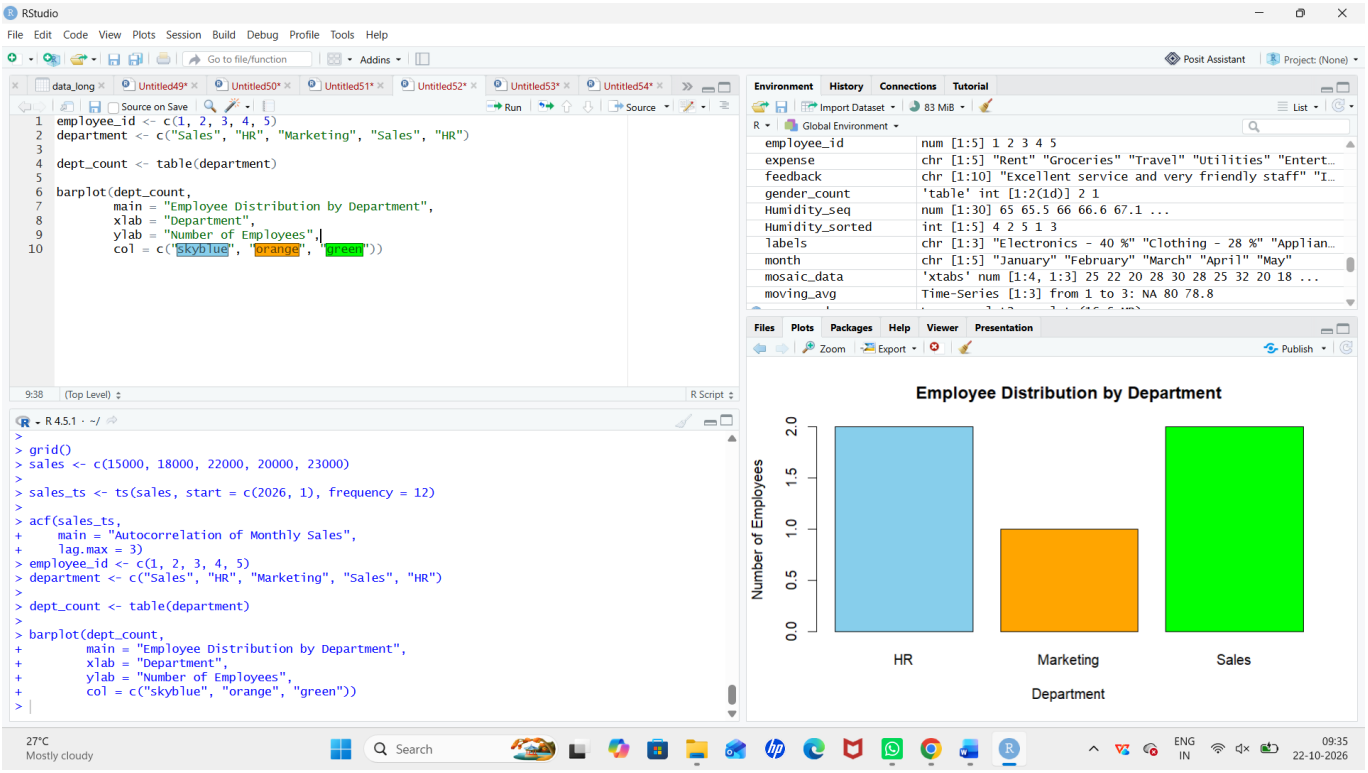
To analyze employee performance data using R and Tableau by creating a department-wise bar chart, a line chart comparing employee performance scores, an interactive dashboard showing department distribution and the relationship between years of service and performance scores, and a table displaying employee details.

1.

```
employee_id <- c(1, 2, 3, 4, 5)
department <- c("Sales", "HR", "Marketing", "Sales", "HR")

dept_count <- table(department)

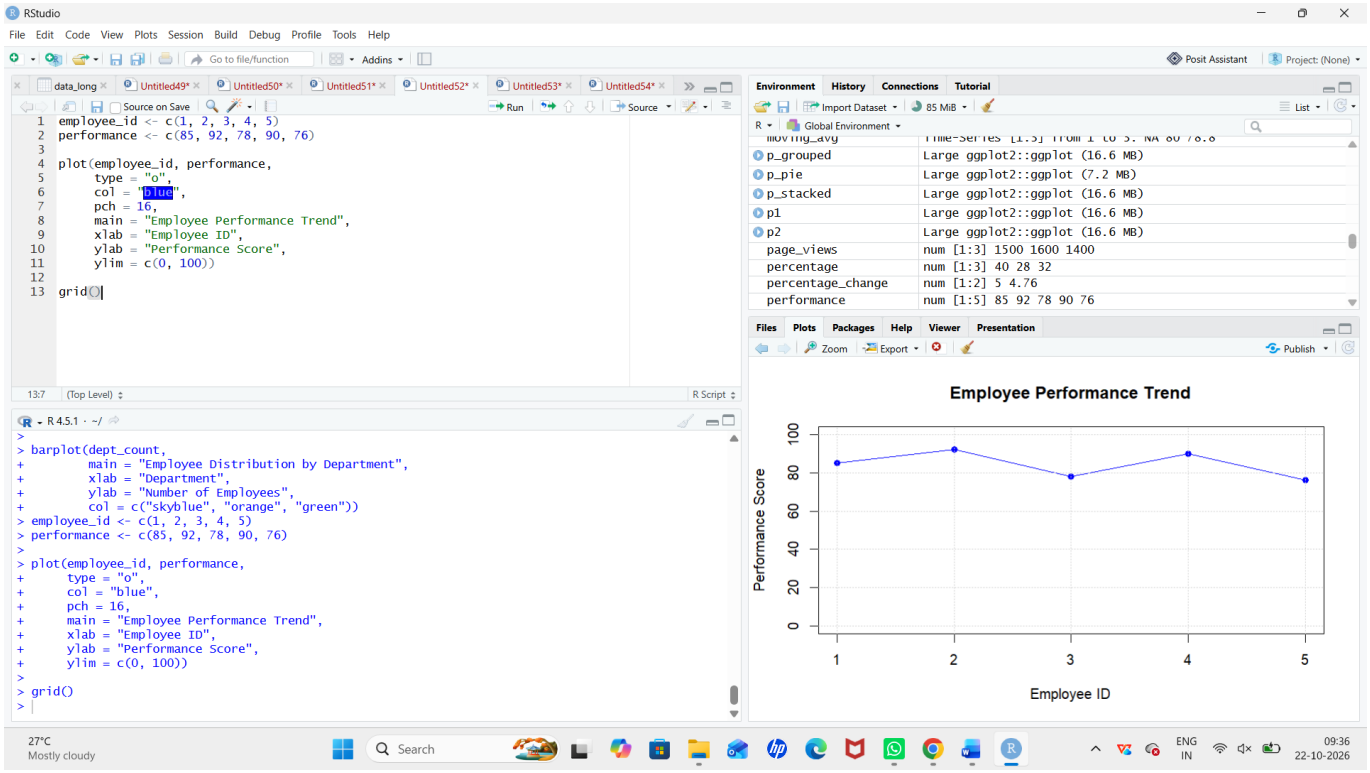
barplot(dept_count,
        main = "Employee Distribution by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = c("skyblue", "orange", "green"))
```



2

```
employee_id <- c(1, 2, 3, 4, 5)
performance <- c(85, 92, 78, 90, 76)

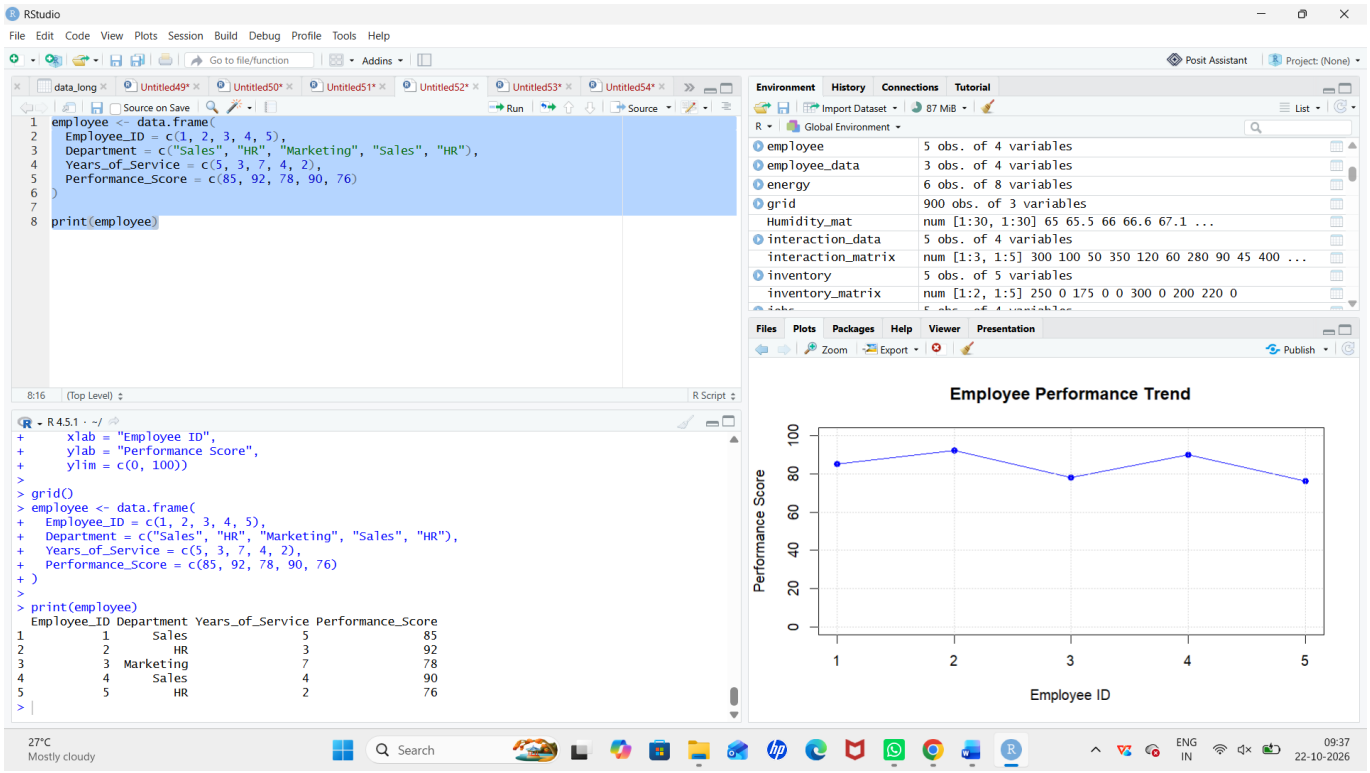
plot(employee_id, performance,
     type = "o",
     col = "blue",
     pch = 16,
     main = "Employee Performance Trend",
     xlab = "Employee ID",
     ylab = "Performance Score",
     ylim = c(0, 100))
```



4.

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee)
```



3.

- Create the employee dataset in Excel or CSV format.
- Open Tableau and connect to the dataset.
- Create Sheet 1 – Department Distribution by dragging Department to Columns and Employee ID to Rows. Change Employee ID aggregation to Count.
- Select the Bar chart visualization.
- Create Sheet 2 – Performance Scatter Plot by dragging Years of Service to Columns and Performance Score to Rows.
- Drag Employee ID to Detail to display each employee as a separate point.
- Add Department to Color to distinguish departments.
- Create a new dashboard and add both sheets.
- Add suitable titles, labels, and colors.

1. Create the employee dataset containing Employee ID, Department, Years of Service, and Performance Score.

2. Define the dataset in R using a data frame.

3. Create a bar chart to visualize the distribution of employees across departments.

4. Generate a line chart to compare performance scores among employees.

5. Create a table in R to display the performance details of each employee.

6. Import the dataset into Tableau.

7. Create a bar chart showing the number of employees in each department.

8. Create a scatter plot to analyze the relationship between years of service and performance scores.

9. Combine both charts into a single Tableau dashboard and add suitable titles and labels.

10. Observe and interpret the visualizations to understand employee distribution and performance differences.

Result

Employee performance data was successfully analyzed using R and Tableau. The bar chart displayed department-wise employee distribution, the line chart compared employee performance scores, and the table presented individual employee details. The Tableau dashboard visualized department distribution and the relationship between years of service and performance scores.